

Report Week 5

For this assignment I worked with 4 datasets. Iris, Automobile MPG, Student Performance and Car Evaluation.

GitHub: <https://github.com/vitali-karmanov/CS450-BYUI/tree/master/week05>

Iris

5% for test and 5 of max-depth:

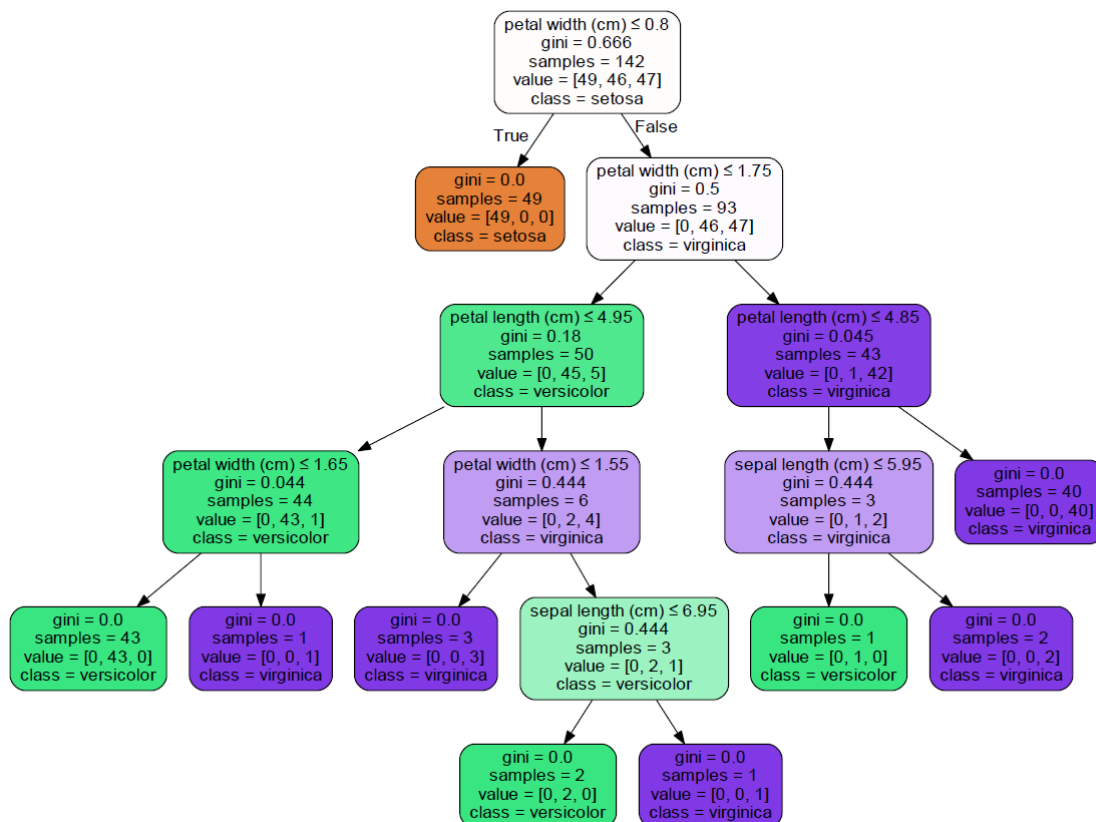
Predicted:

[1 1 0 1 2 1 2 2]

Actual:

[1 1 0 1 2 1 2 2]

Accuracy is 100.0



Bigger version in GitHub

5% for test and 2 of max-depth:

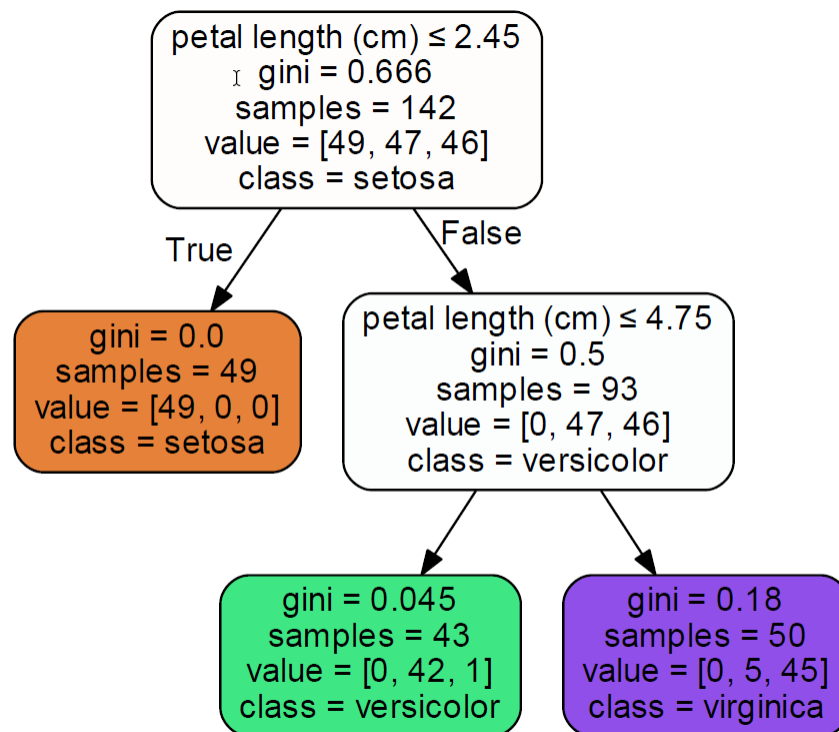
Predicted:

[0 2 1 2 2 2 1 2]

Actual:

[0 2 1 2 2 2 1 1]

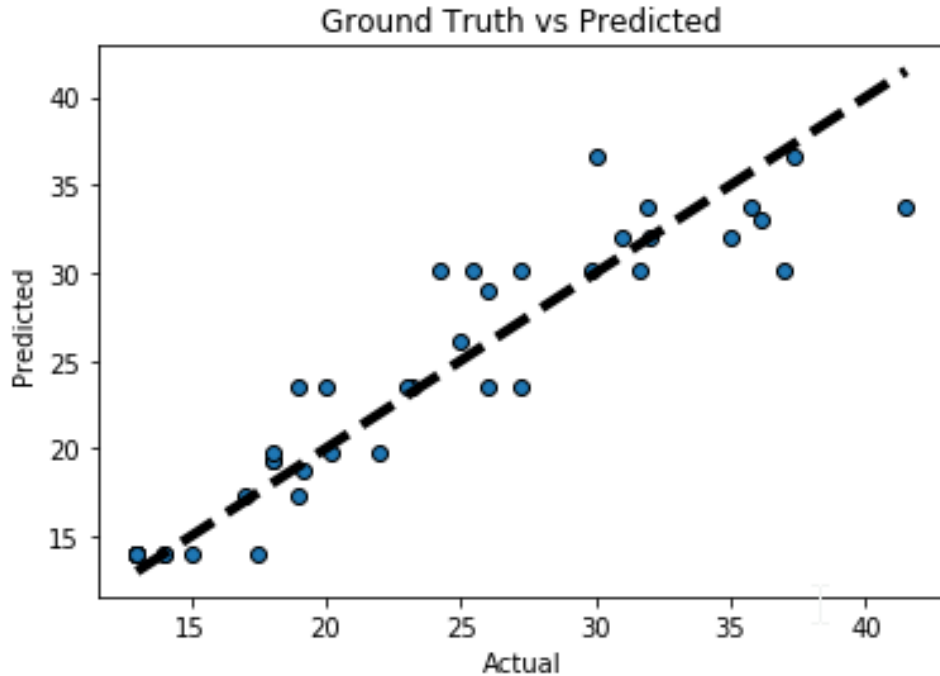
Accuracy is 87.5



Notes: Less effective with less depth. Classification.

Automobile MPG

10% for test and 5 of max-depth:



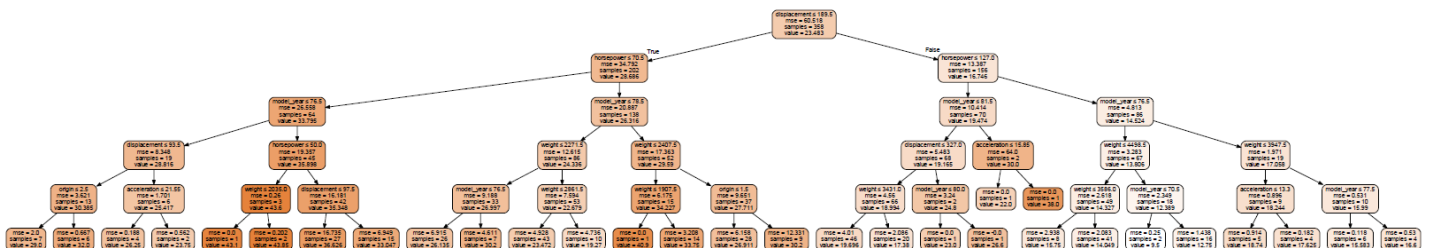
Predicted:

```
[30.2      30.2      14.04878049 14.04878049 19.69565217 23.47209302
14.04878049 33.75      19.69565217 23.47209302 14.04878049 23.47209302
23.47209302 30.2      30.2      23.47209302 33.04666667 17.38
30.2      14.04878049 32.      32.      23.47209302 29.
26.13461538 36.62592593 33.75      14.04878049 19.27      19.27
19.69565217 30.2      14.04878049 14.04878049 14.04878049 36.62592593
17.38      18.74      33.75      32.      ]
```

Actual:

```
[37.  27.2 14.  13.  22.  27.2 15.  41.5 20.2 26.  14.  20.  19.  25.4
29.8 23.2 36.1 19.  31.6 13.  35.  31.  23.  26.  25.  30.  31.9 13.
18.  18.  18.  24.2 13.  17.5 13.  37.3 17.  19.2 35.7 32. ]
```

Automobile MPG - Accuracy: 86.71335237059925%

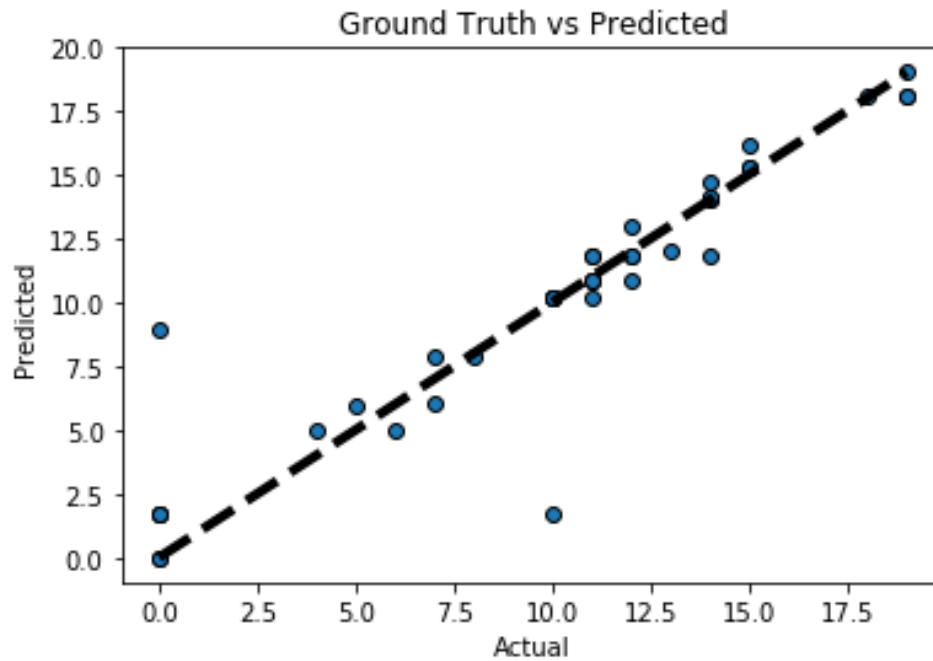


Bigger version in GitHub

Notes: Handling missive data, regression classifier, numerical data.

Student Performance

10% for test and 5 of max-depth:



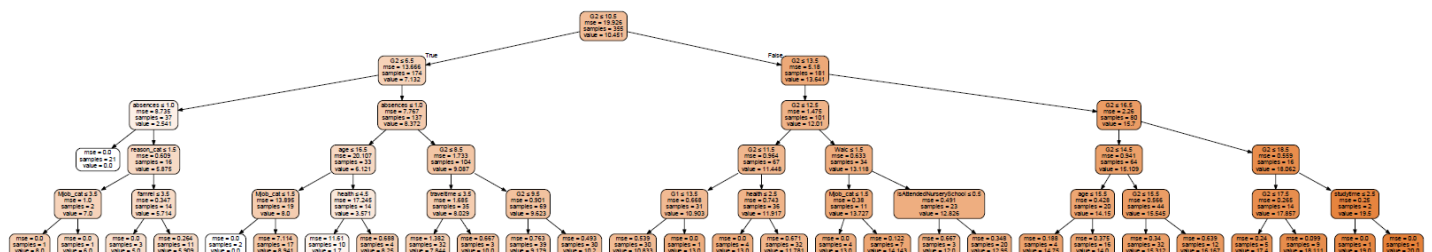
Predicted:

```
[10.2      7.84375    11.78125    14.         5.90909091  1.7
  1.7      16.16666667 11.78125     1.7         1.7         10.83333333
 18.11111111 12.95      7.84375    10.2         15.3125     14.
 11.78125    10.2       5.         12.         11.78125     5.
 15.3125     0.         10.2       14.75        10.83333333 10.83333333
 11.78125    10.83333333 18.11111111 14.14285714 10.2         19.
 6.         18.11111111 8.94117647 0.         ]
```

Actual:

```
[10  8 12 14  5  0 10 15 11  0  0 11 19 12  7 11 15 14 11 10  6 13 14  4
 15  0 10 14 12 11 12 11 19 14 10 19  7 18  0  0]
```

Student Performance - Accuracy: 85.21437246491223%



Bigger version in GitHub

Notes: Regression classifier, one-hot encoding, numerical data.

Car Evaluation

2% for test and 5 of max-depth:

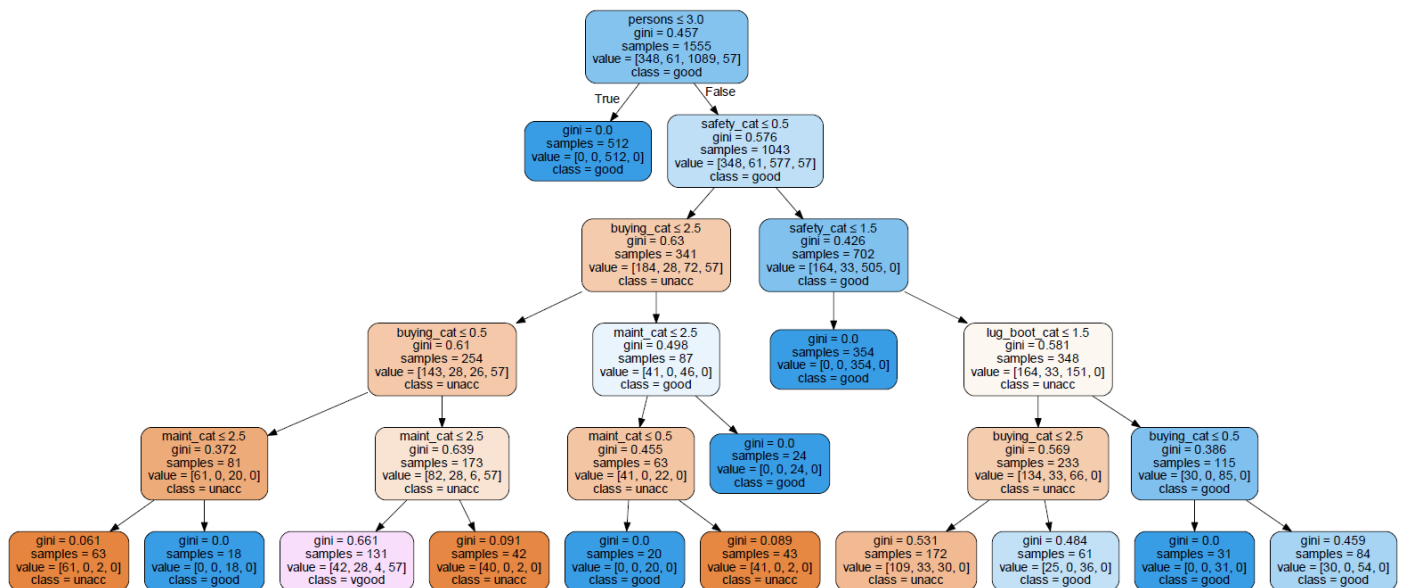
Predicted:

```
[ 'acc' 'acc' 'unacc' 'vgood' 'acc' 'unacc' 'unacc' 'unacc' 'acc' 'unacc'
  'unacc' 'unacc' 'vgood' 'unacc' 'unacc' 'unacc' 'acc' 'unacc' 'acc'
  'unacc' 'unacc' 'unacc' 'unacc' 'unacc' 'acc' 'unacc' 'acc' 'acc' 'acc'
  'unacc' 'unacc' 'unacc' 'unacc' 'unacc' 'unacc' ]
```

Actual:

```
[ 'acc' 'good' 'unacc' 'vgood' 'acc' 'unacc' 'unacc' 'unacc' 'acc' 'unacc'
  'unacc' 'acc' 'vgood' 'unacc' 'unacc' 'unacc' 'acc' 'unacc' 'unacc'
  'unacc' 'unacc' 'unacc' 'unacc' 'unacc' 'acc' 'unacc' 'acc' 'acc' 'acc'
  'unacc' 'unacc' 'unacc' 'unacc' 'unacc' 'unacc' ]
```

Car Evaluation - Accuracy: 91.42857142857143%



Bigger version in GitHub

Notes: Regression classifier, label encoding, numerical data.

Final Notes:

I believe that my assignment goes beyond the requirements. Because I implemented 4 datasets, I used Numerical Data, Categorical data (label and one-hot encoding), Missing data, and using Regression and Classification.

In the program you can choose to specify the percentage of test data, and the depth of the tree.

Finally, the visual representation of a tree, is dynamically created and for regression I'm also showing the plots.